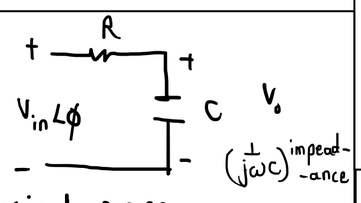
5. All kinds of stuff

28/03/25 9:47 PM

Source has only one frequency :



$\left(\frac{1}{j\omega C}\right)$ impedance

$$\bar{V}_o = \frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C} + R} \times \bar{V}_{in}\angle\phi$$

$\frac{1}{j\omega C} \to \angle -90$

$\angle \tan^{-1}\left(\frac{-1}{R\omega C}\right)$

$|\bar{V}_o|$ has physical sense

$|\bar{V}_o| = \frac{1}{2}$ say toh

$|\bar{V}_{in}| = 1$ as $\bar{V}_{in} = \frac{V_i}{\sqrt{2}}(\ )$

$V_{in} = \frac{\sqrt{2}}{\sqrt{2}}\angle\phi'$ & $V_i = \sqrt{2}$ given

steady state:

$$V_o(t) = \sqrt{2} \times \frac{1}{2} \times 1 \times \cos\left(\omega t + \phi_1 - \frac{\pi}{2} - \tan^{-1}\left(\frac{1}{\omega RC}\right)\right)$$

depends on ω

Say input is $V_1\cos(\omega t + \phi_1) + V_2\cos(\omega t + \phi_2)$

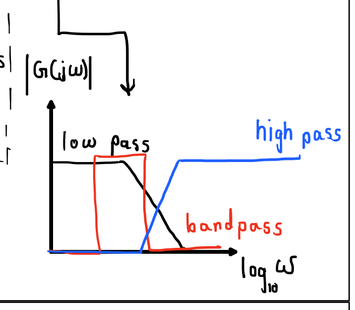
$\bar{V}_1 = \frac{V_1}{\sqrt{2}}\angle\phi_1$ $\quad \bar{V}_2 = \frac{V_2}{\sqrt{2}}\angle\phi_2$

o/p due to $= G(j\omega_1)\cdot\bar{V}_1 + G(j\omega_2)\bar{V}_2$

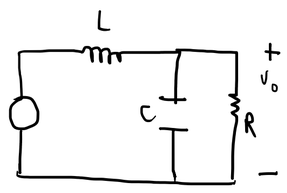
$G(j\omega) \to$ Transfer $f^n \longrightarrow \left(\frac{1}{j\omega CR + 1}\right)$ MS 101 mein jo tha

$V\sin(\omega t) = \frac{V}{\sqrt{2}}\angle -\pi/2$

we use logarithmic scales [Assymptotic plots]



Buck?



$$\left(\frac{\bar{V}_o}{\bar{V}_{in}}\right) = \frac{\left[\frac{\frac{R}{j\omega C}}{R + \frac{1}{j\omega C}}\right]}{\left(\frac{\frac{R}{j\omega C}}{R + \frac{1}{j\omega C}}\right) + j\omega L} = \frac{R^2}{R - \omega^2(LR + j\omega L)}$$

$$\text{Gain} = \left|\frac{\bar{V}_o}{\bar{V}_i}\right| = G(j\omega) = \left|\frac{R}{\sqrt{(R^2 - \omega^2 LR^2 \ldots}}\right|$$

$$\int_0^{2\pi} \sin(t)\cdot\sin(mt)\,dt = 0$$

$m \neq 0, 1$

Basically $\int_0^{2\pi} s(nt)\cdot s(mt)\,dt = 0$ if $m \neq n$ $\forall\, m, n$'s

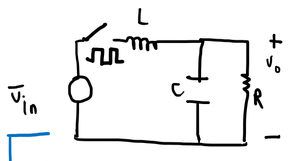
$\sin(mt - \phi)$ hua toh bhi $= 0$ ✓✓

$\frac{e^{i(mt-\phi)} - e^{-i(mt-\phi)}}{2}$

$\int s(n\omega t)\cdot s(m\omega t)\,dt = 0$

$m\omega t - \phi$ ho toh bhi it works

$D_{sec} \to ON$; $(1-D)_{sec} \to OFF$



Basically high V DC is converted to low voltage DC

BUCK CONVERTER

P_L & P_C $= 0$ So best

cuz R hi load hai toh uske thru toh pwr loss nhi hai kinda aise bol skte hai

Fourier Baba :(

$$\underset{\text{periodic } f^n}{f(t)} = \sum A_n \sin(n\omega t) + \sum B_m \sin(m\omega t) + a_0$$

Basically × by $\sin(n\omega t)$ for $n = 1$ se . . . . to get A_n's cuz baaki terms 0 hojaenge

$$\int_0^T f(t) = \cancel{\int \Sigma} + \cancel{\int \Sigma} + \int_0^T a_0\,dt = a_0 T$$

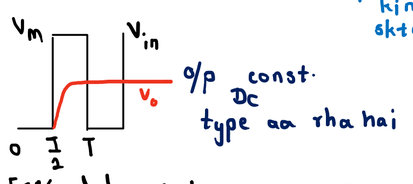
$$\left[a_0 = \frac{1}{T}\int_0^T f(t)\right] \quad \& \quad \left[A_n = \frac{2}{T}\int_0^T f(t)\cdot\sin(n\omega t)\,dt\right]$$

$0 \to T$ not necessary, just 1 full time period ka interval chahiye

$-\frac{T}{2} \to \frac{T}{2}$ works

$$\left[B_n = \frac{2}{T}\int_T f(t)\cdot\cos(n\omega t)\,dt\right]$$

if input is a sq wave toh we can write it as Σ sin waves



o/p const. DC type aa rha hai

50% duty cycle

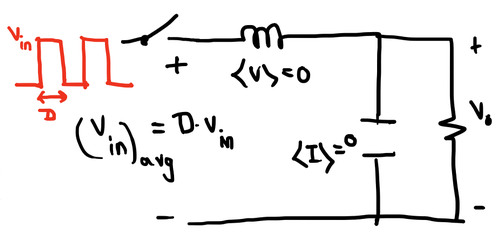
If switching f↓ toh o/p could have ↑ripples

$$\left[V_{avg} = D V_{in}\right]$$

Duty cycle

P_{avg} thru L and C is 0 [Clearly]

[For inductor in steady state avg voltage = 0
For Capacitor in steady state avg current = 0]



$\langle v\rangle = \langle v_L\rangle^{0} + \langle v_o\rangle$

$$\left[DV_{in} = \langle V_o\rangle\right]$$